



MAX PLANCK INSTITUTE
FOR MULTIDISCIPLINARY SCIENCES

Computational Electrophysiology of Potassium and Chloride Channels



Wojciech Kopec^{1,2}, Chenggong Hui², Reinier de Vries², Edward Mendez-Otalvaro², Ivan Manóza¹, Jakub Mróz³, Artur Góra³, Bert L. de Groot²
w.kopec@qmul.ac.uk

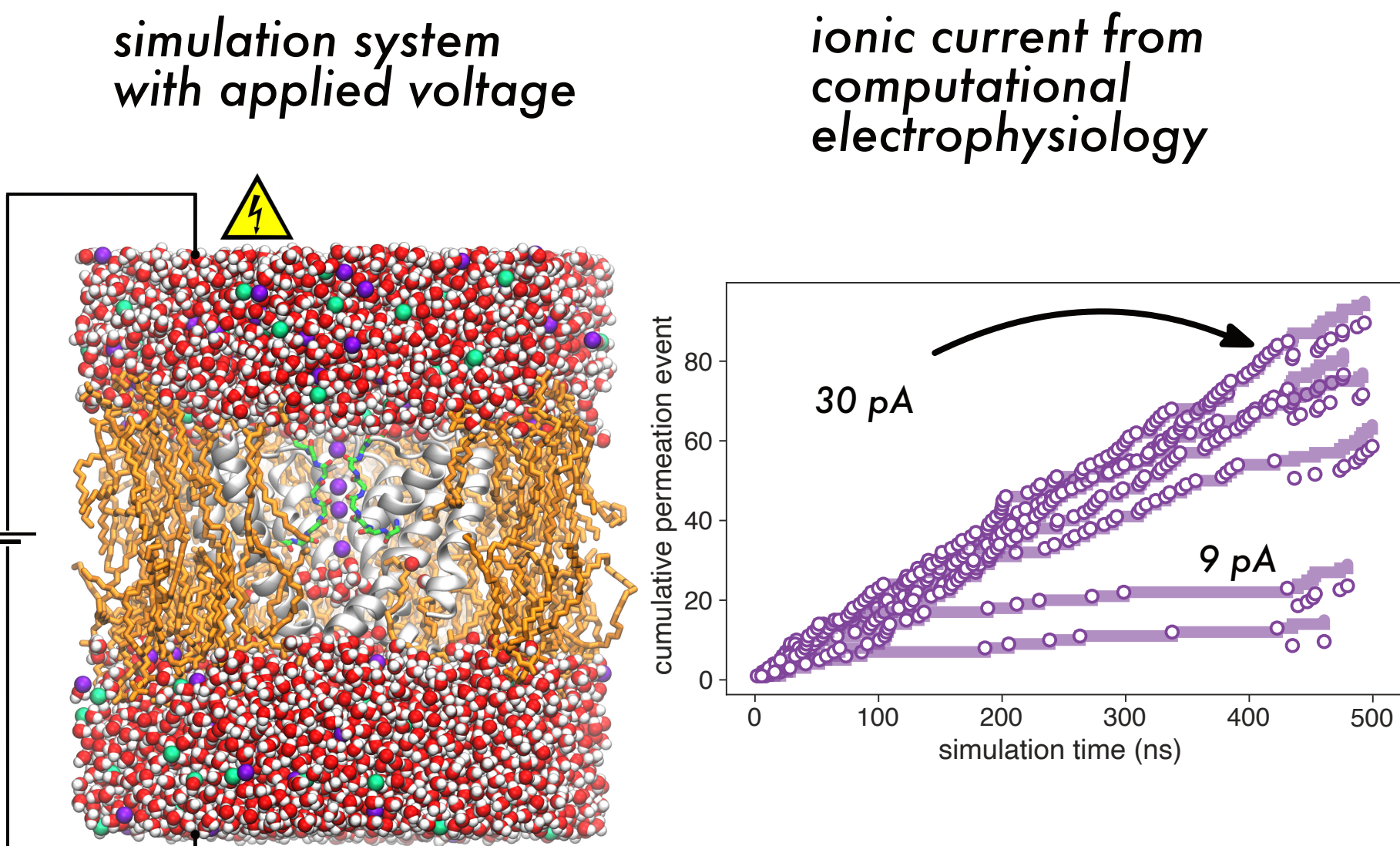
¹Department of Chemistry, Queen Mary University of London, 327 Mile End Road, E1 4NS London, UK

²Computational Biomolecular Dynamics Group, Max Planck Institute for Multidisciplinary Sciences, Am Fassberg 11, 37077 Göttingen, Germany

³Tunneling Group, Biotechnology Centre, ul. Krzywoustego 8, Silesian University of Technology, 44-100 Gliwice, Poland

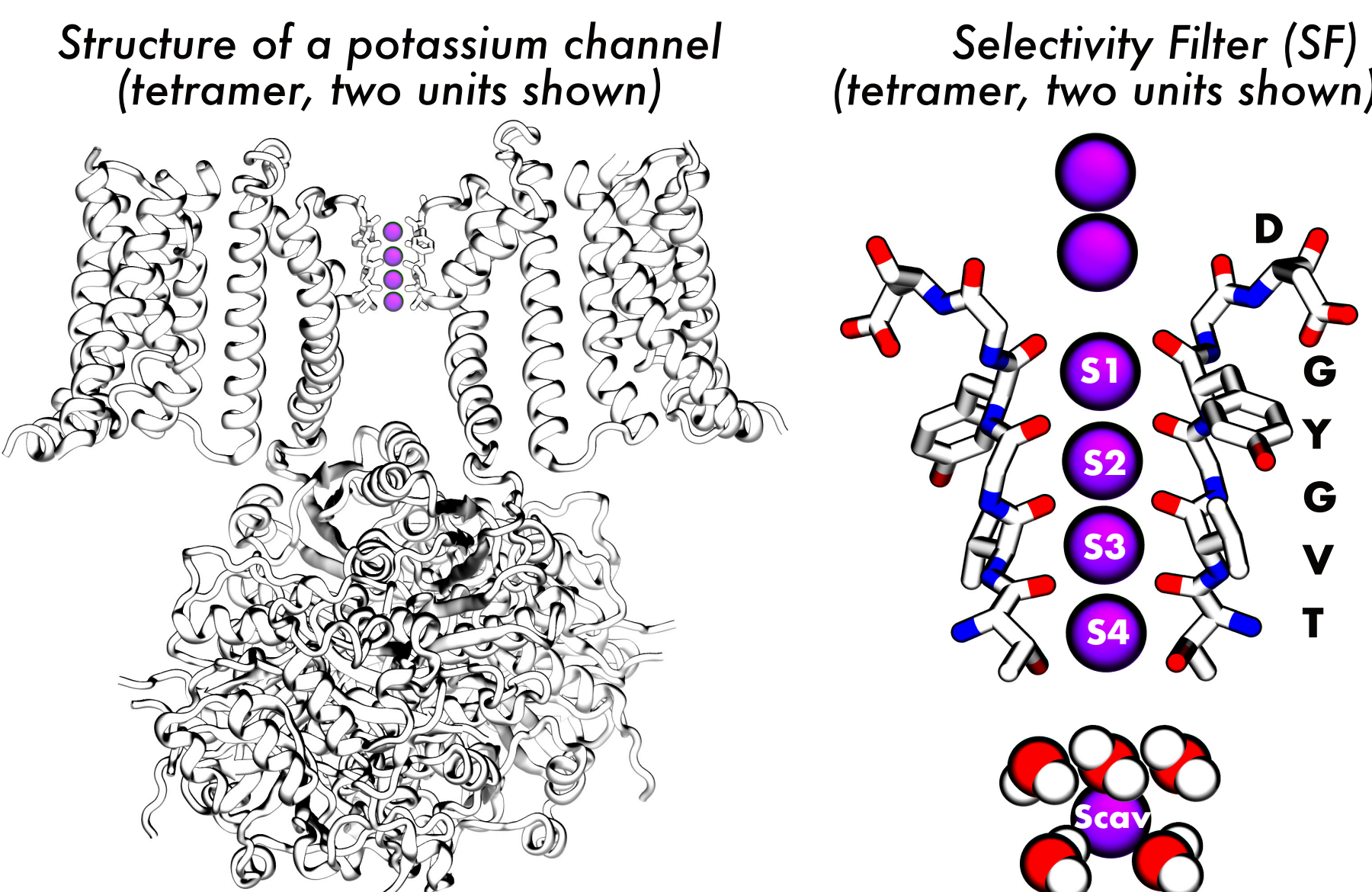
Introduction

We use Molecular Dynamics (MD) based, Computational Electrophysiology (Comp. El.) simulations to obtain a clear view into ion permeation on the atomistic scale. We start from high resolution structures of potassium channels and apply an electric field (membrane voltage), akin to electrophysiological measurements, that leads to individual ion permeation events. Counting these events results in measured ionic currents.

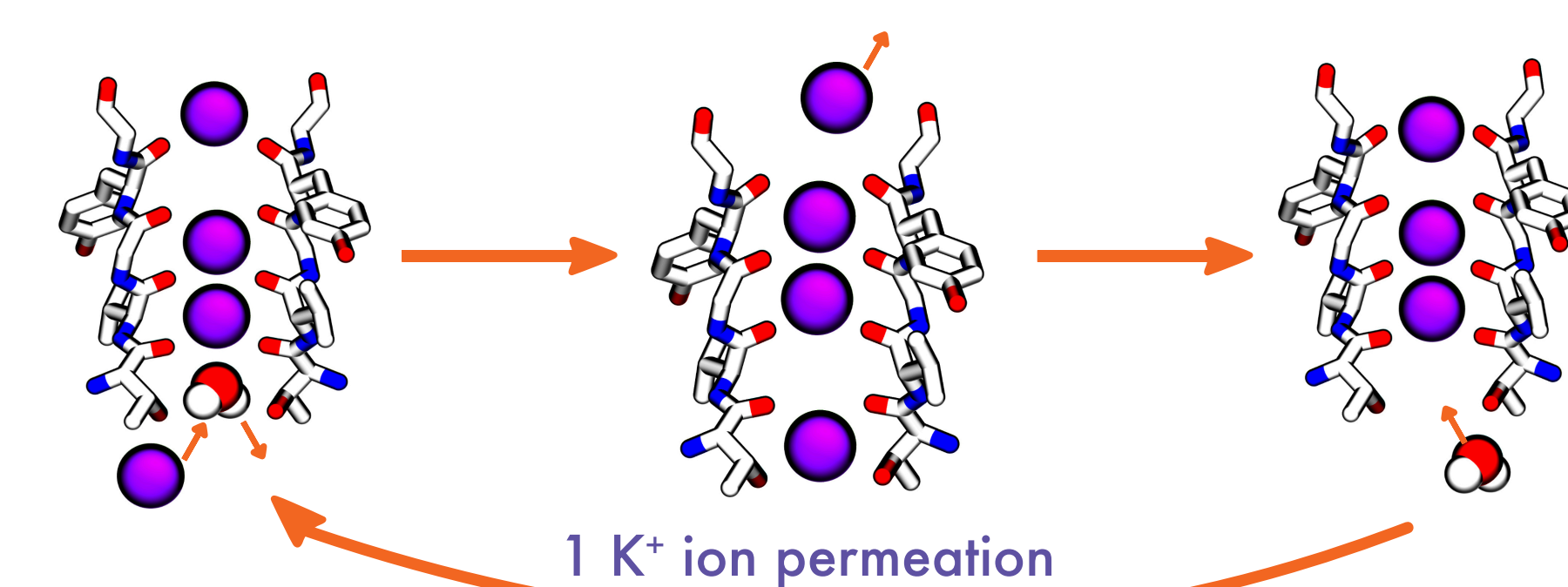


Potassium permeation in K⁺ channels

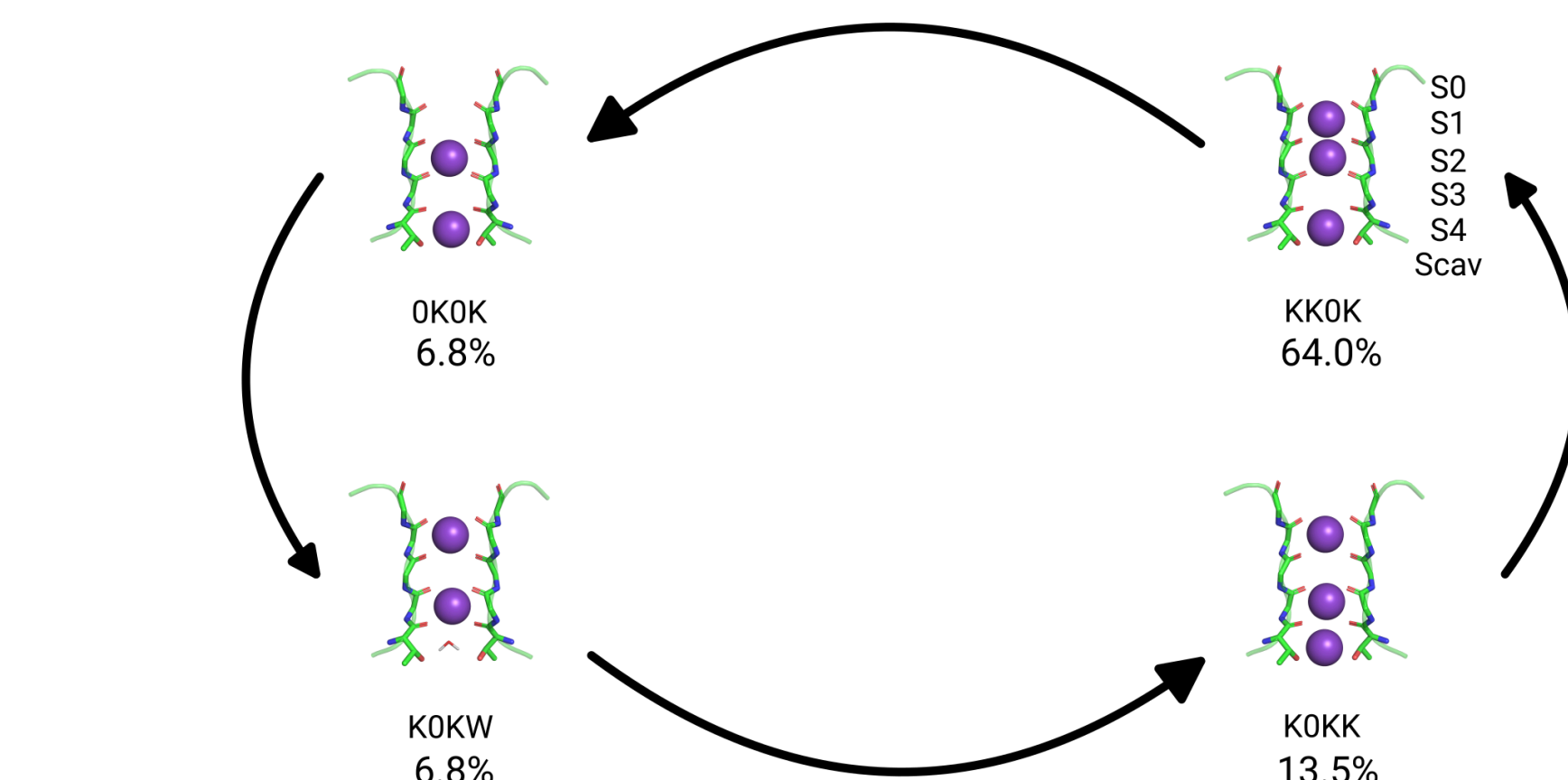
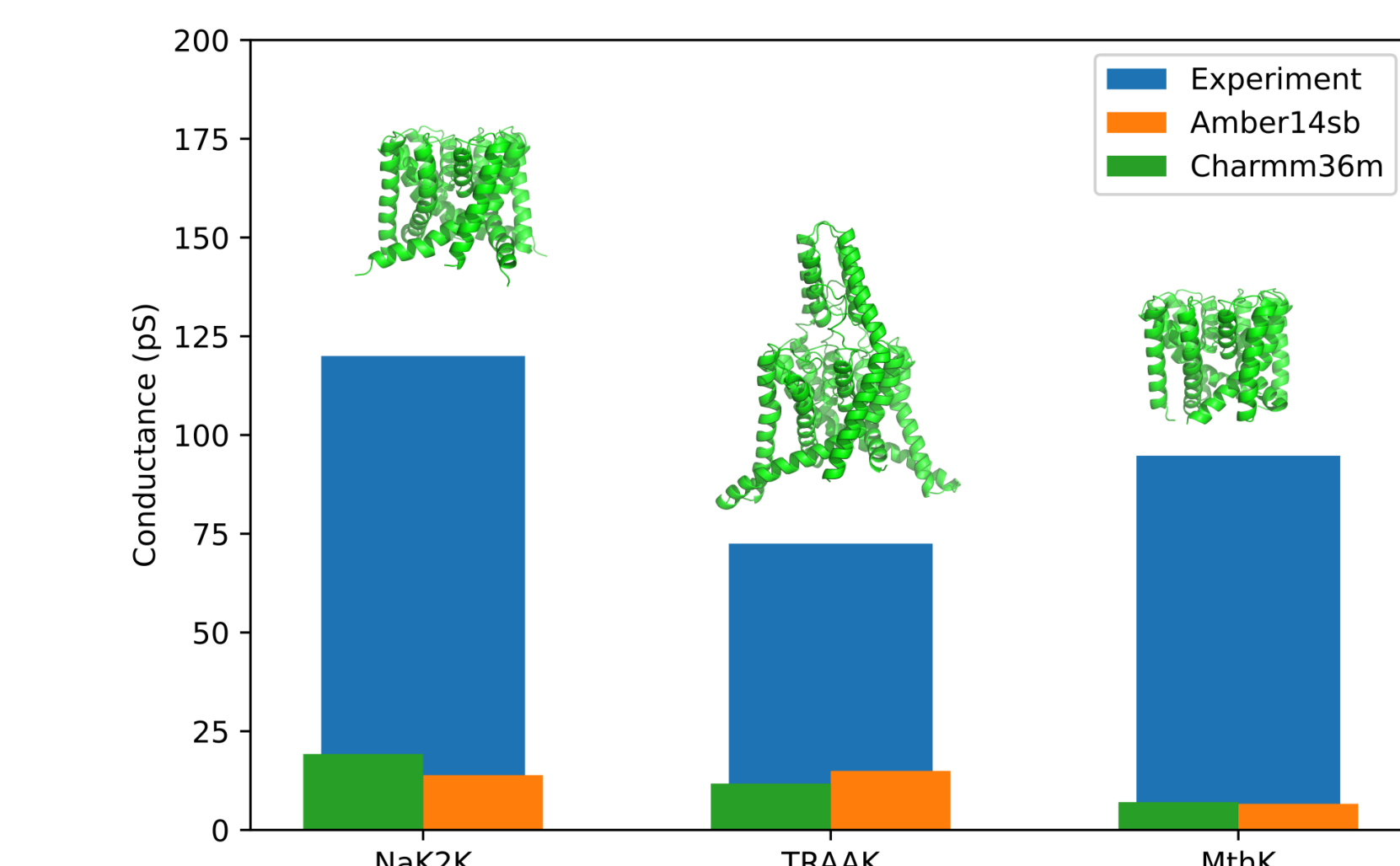
Potassium channels facilitate the passage of **K⁺ ions** through membranes to near-diffusion limited rates. The main structural element of potassium channels - **the selectivity filter** - is conserved in all canonical potassium channels, and yet individual channels can vary in their conduction rates by at least one order of magnitude.



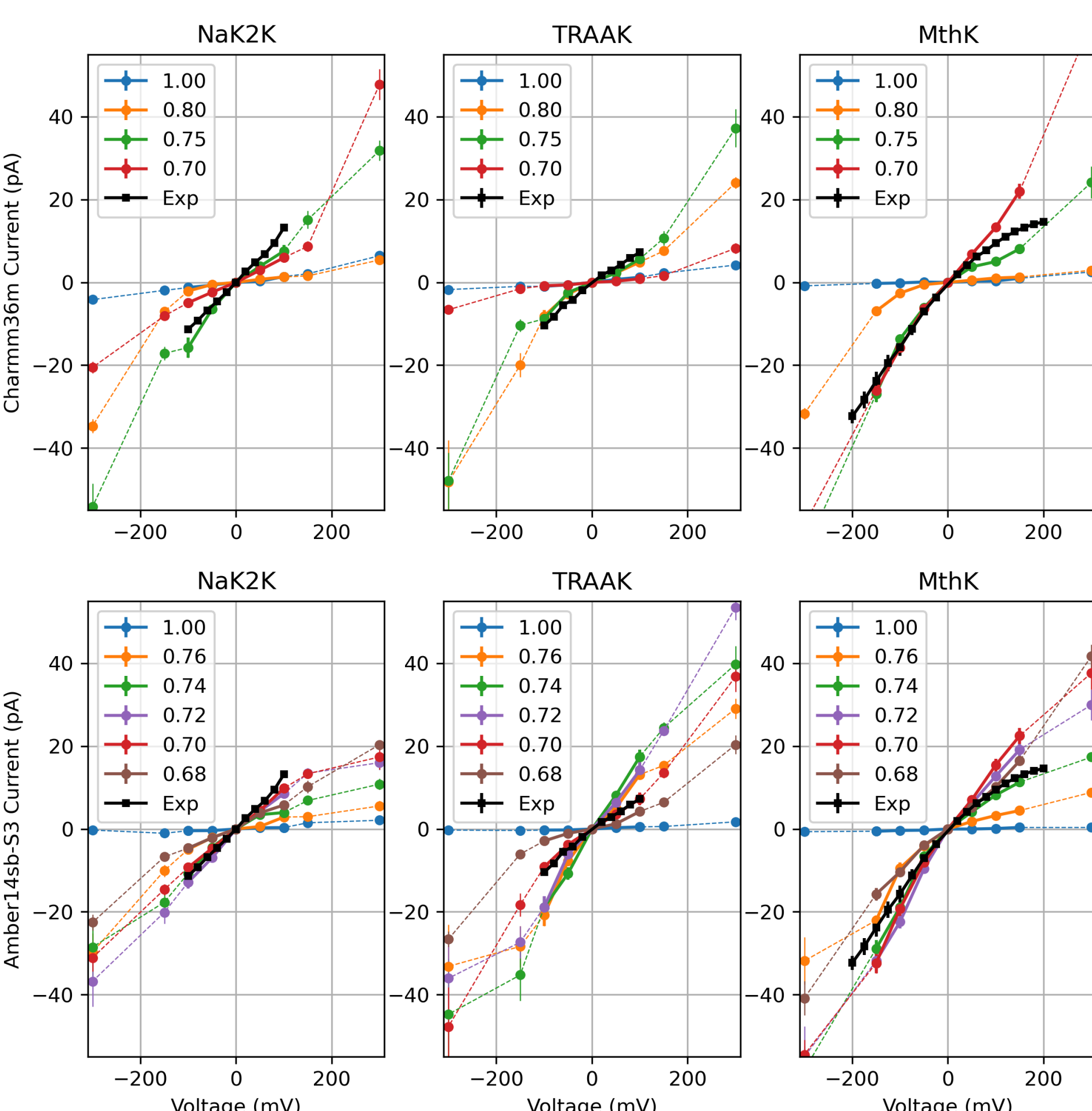
Recent combined experimental and computational studies [1,2] suggest that potassium ions permeate through the channel via the so-called 'direct knock-on' mechanism.



However, the single channel conductance is typically underestimated in simulations.



We developed new ion parameter sets that bring simulated currents close to experimental ones and allow us to simulate the whole I-V curves.

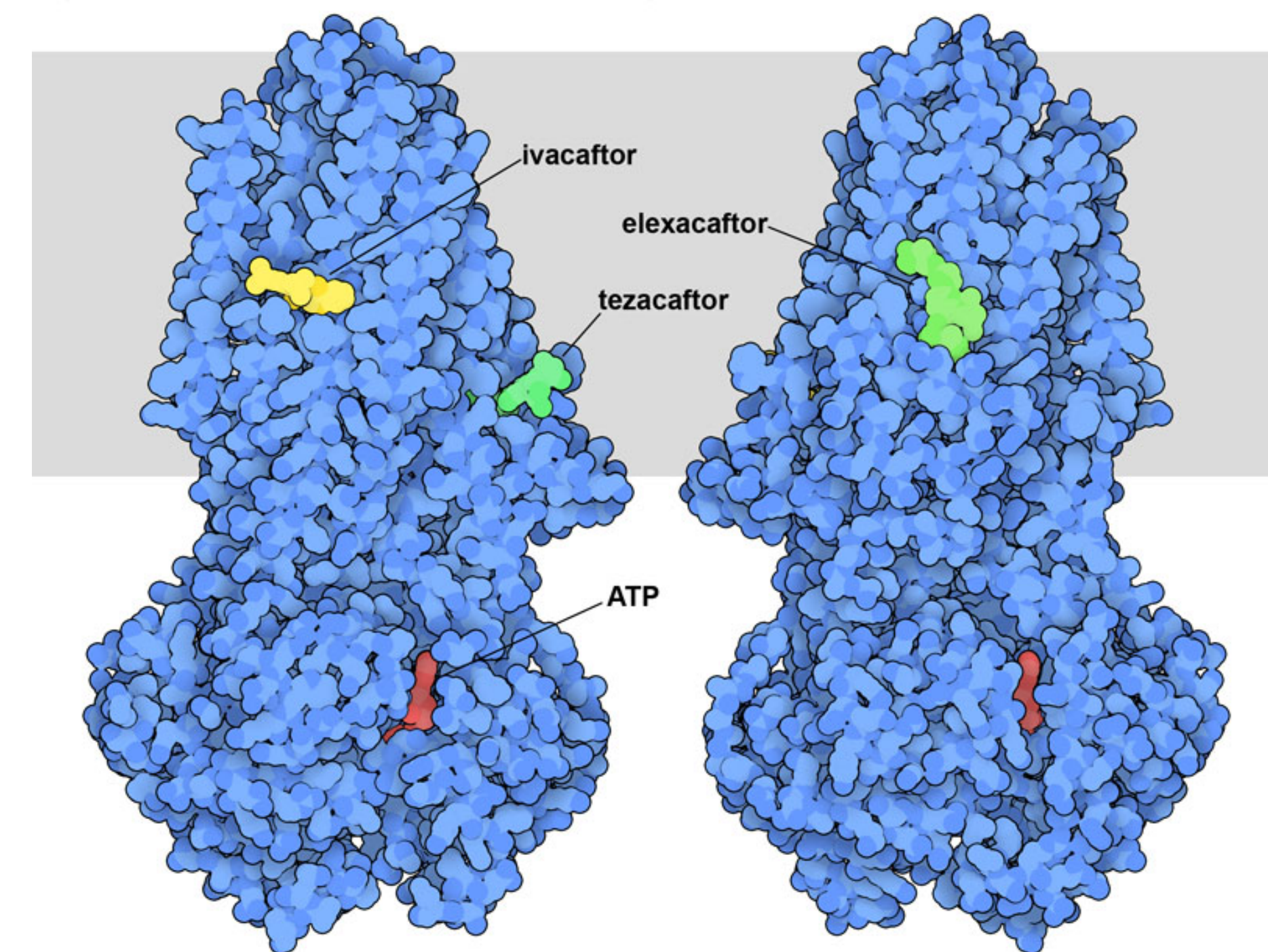


Hui, de Vries, Kopec, de Groot, 'Charge scaling in potassium channel simulations: implications for conductance, ion occupancy, voltage response and selectivity' *in preparation*.

Chloride conduction through CFTR

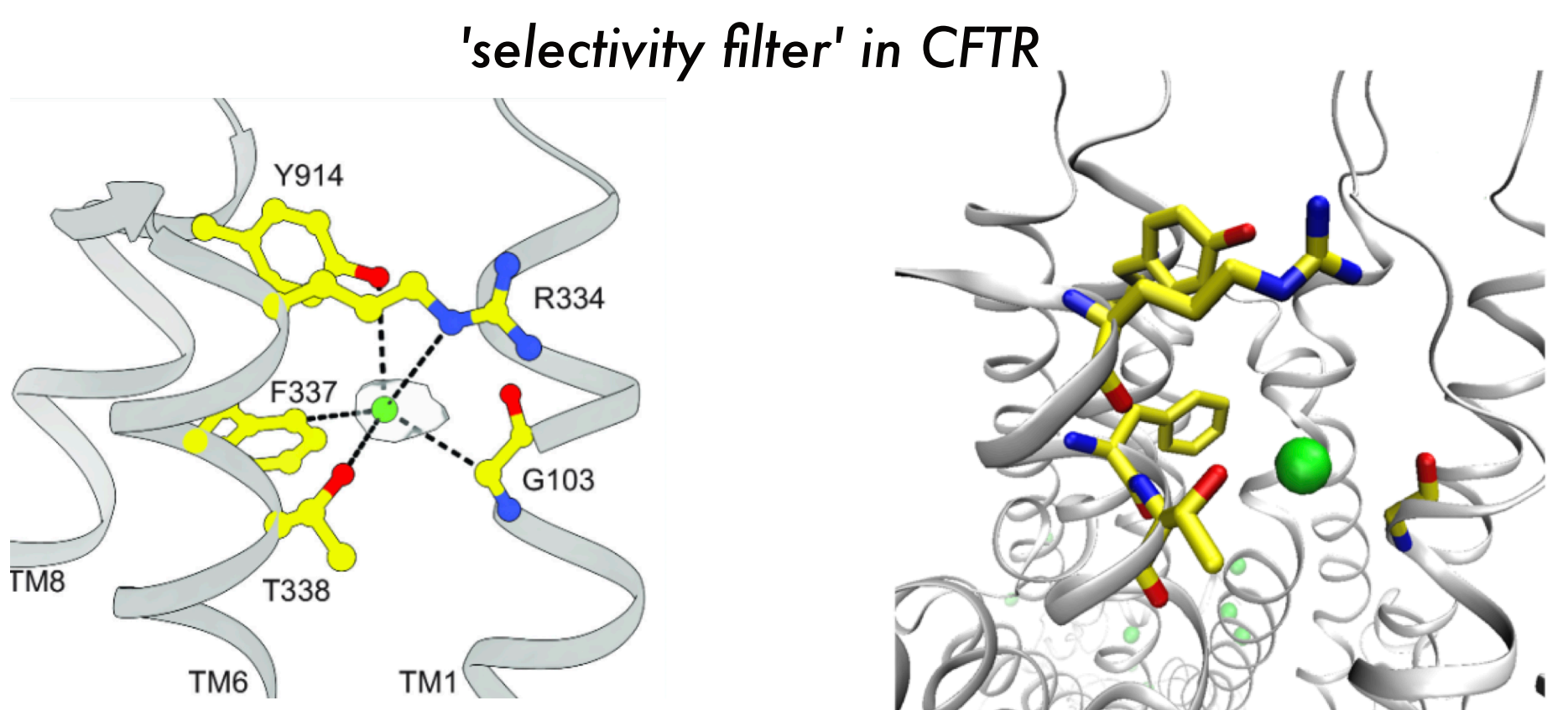
Cystic fibrosis transmembrane conductance regulator (CFTR): an ABC transporter-class ion channel that conduct chloride and bicarbonate ions across epithelial cell membranes. Nearly 1000 cystic fibrosis mutations have been described. Several drugs enhancing CFTR functions have been proposed [5].

CFTR with three drugs used to treat cystic fibrosis

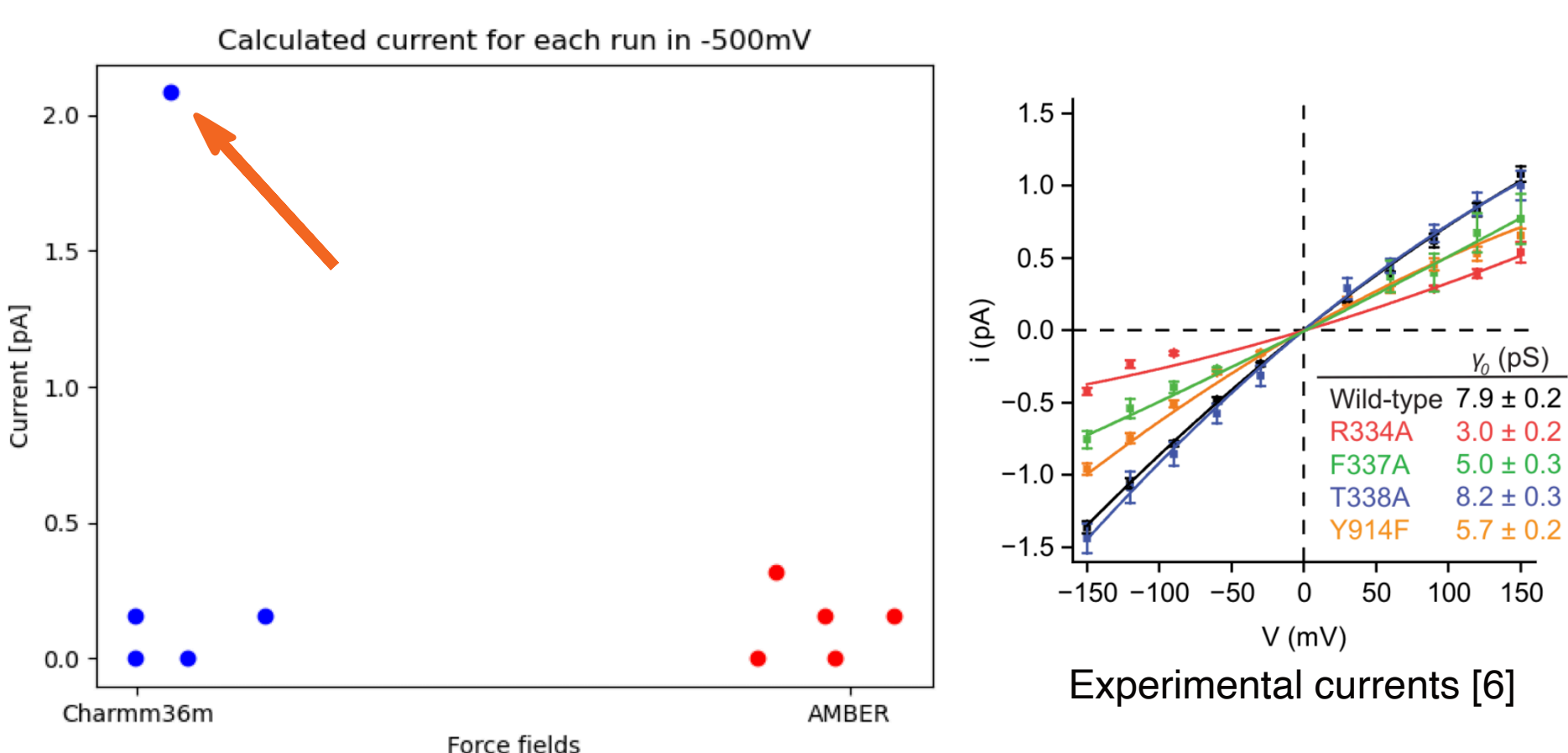


<https://pdb101.rcsb.org/motm/293>

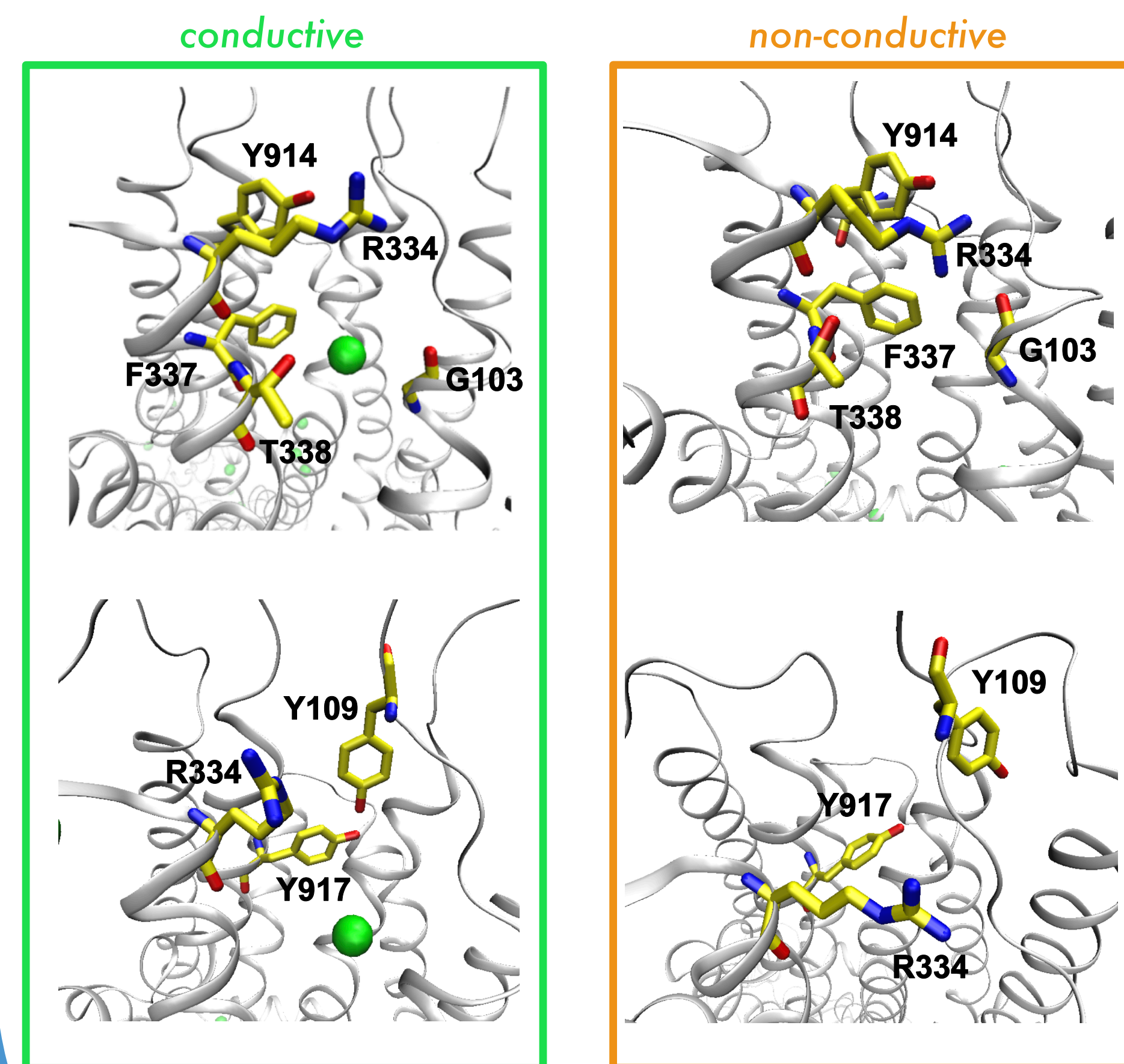
We used Computational Electrophysiology simulations to study chloride permeation. In our simulations, Cl⁻ transiently binds to the same region as in cryoEM structures [6].



So far, we recorded one highly conductive simulation.



High conduction was enabled by a rearrangement of several side chains in the bottleneck region ('selectivity filter').

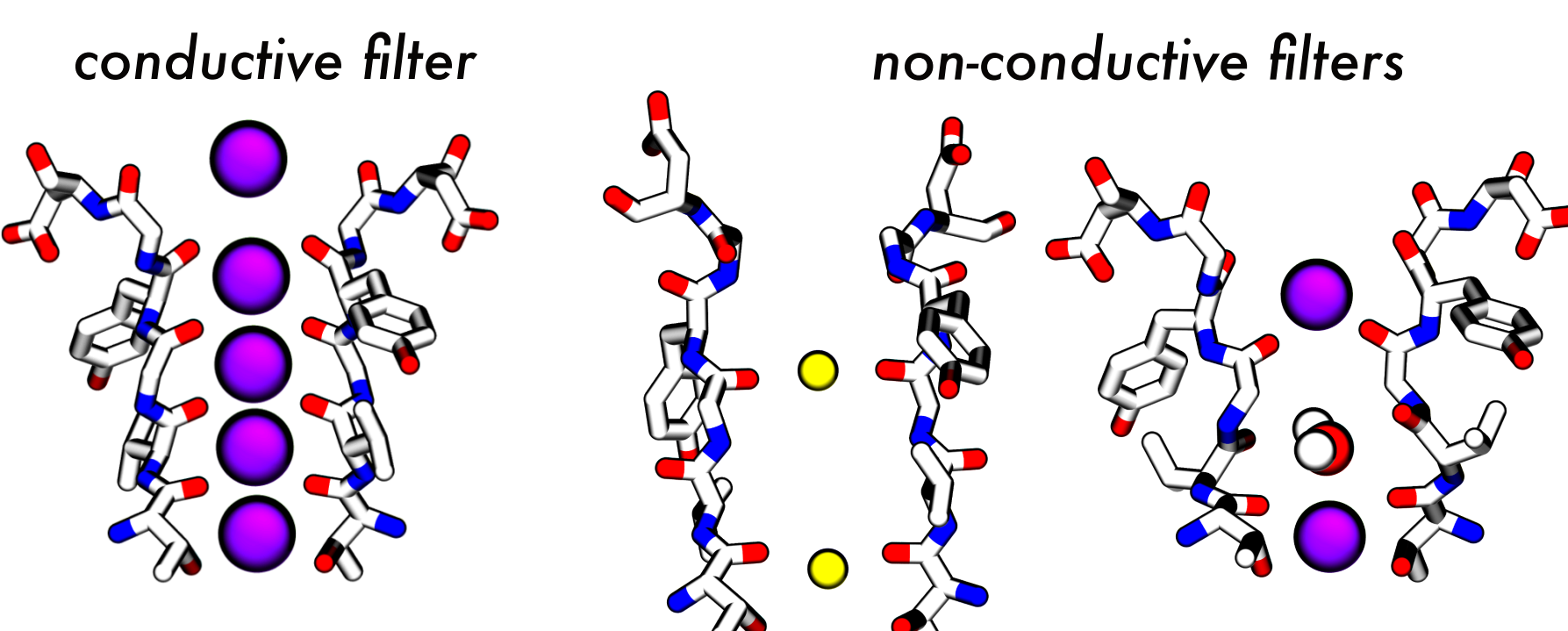


References

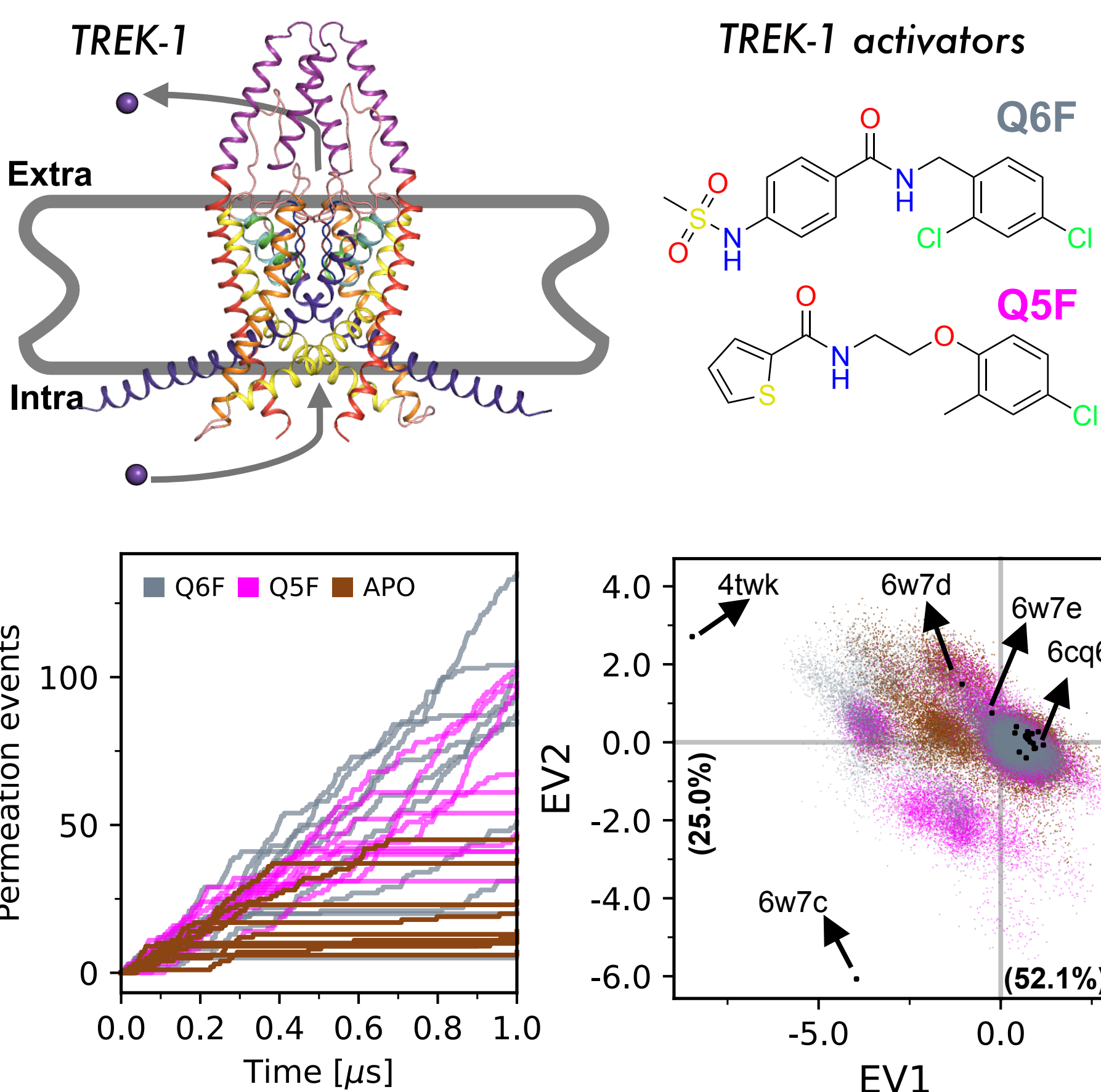
- [1] Köpfer D., Song C., Gruene T., Sheldrick G. M., Zachariae U., de Groot B. L., *Science* 346(6207): 352, 2014
- [2] Öster C., Hendriks K., Kopec W., Chevelkov V., Shi C., Michl D., Lange S., Sun H., de Groot B. L., Lange A., *Science Advances* 5(7): eaaw6756, 2019
- [3] Panyi G., Deutsch C., *Journal of General Physiology* 128, 547, 2006
- [4] Lolicato M., Arrigoni C., Mori T., Sekioka Y., Bryant C., Clark K. A., Minor, D. L. Jr. *Nature* 2017.
- [5] Gramegna A., Contarini M., Aliberti S., Casciaro R., Blasi F., Castellani C., *Intr. J. Mol. Sci.* 2020.
- [6] Levering J., Chen J., *PNAS* 2024.

Selectivity Filter gating in TREK-1

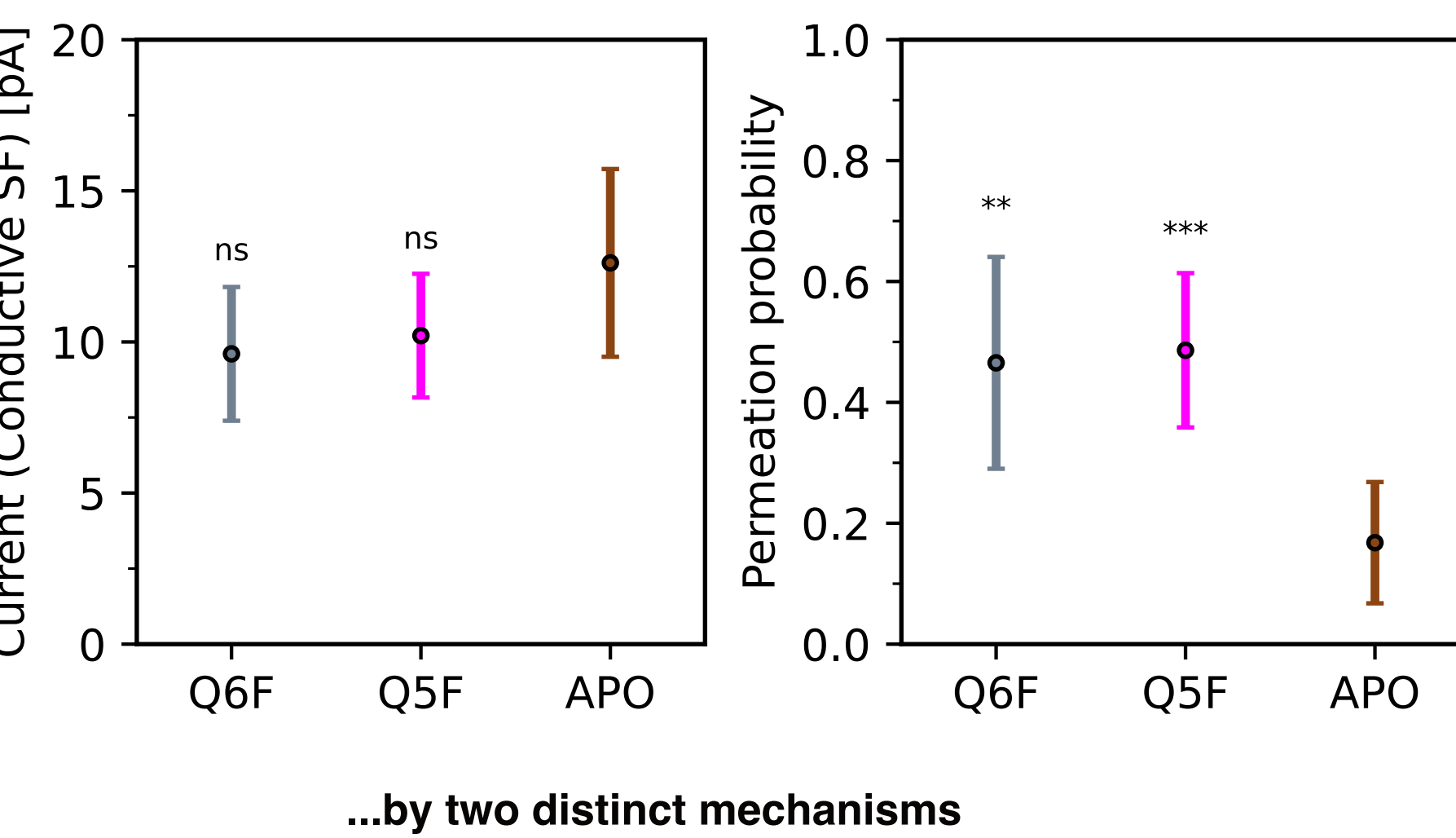
The selectivity filter (SF) of many potassium channels undergoes conformational transitions that allow or halt ion permeation in a process called inactivation or selectivity filter gating [3].



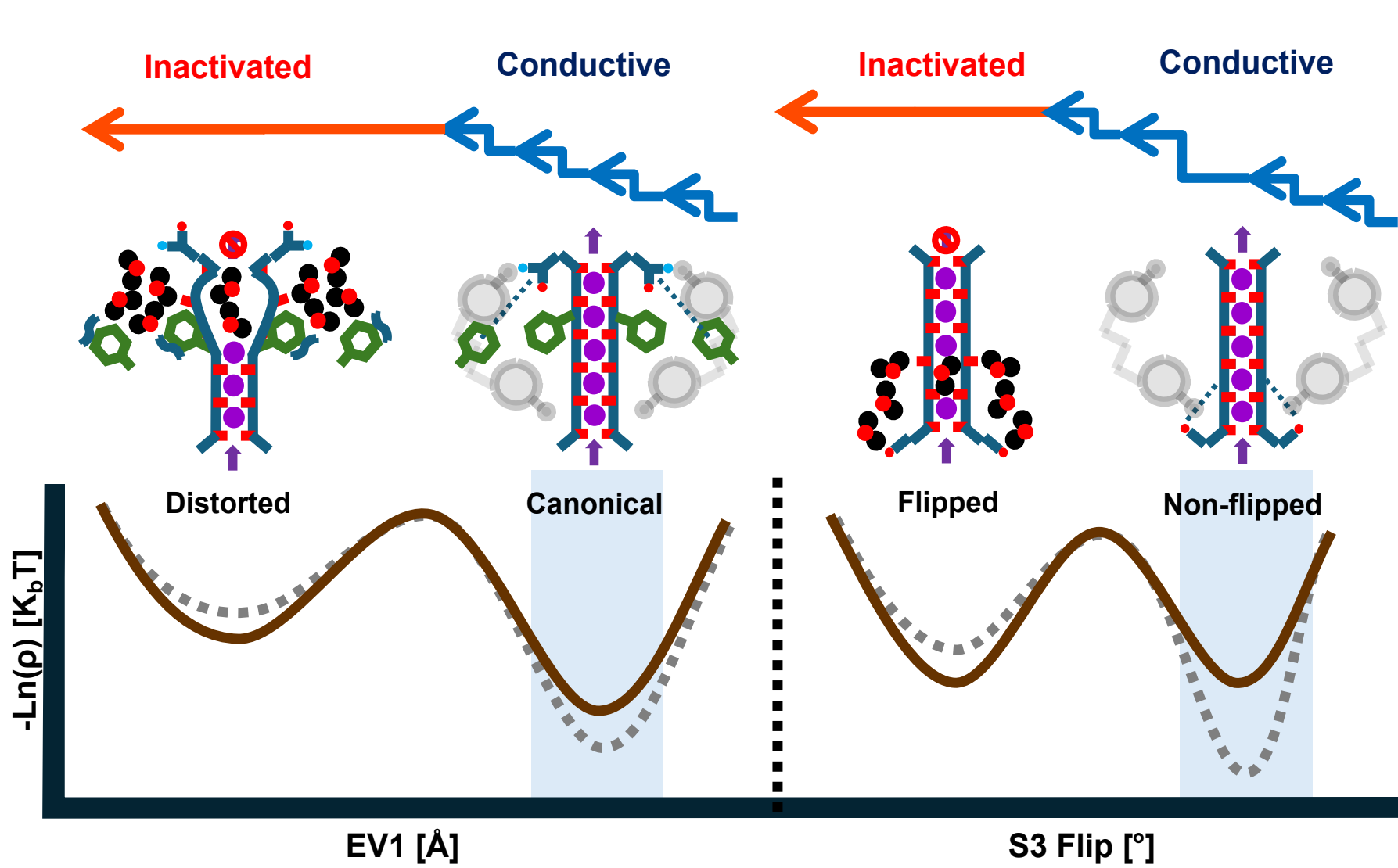
We set out to investigate the effect of two activators (Q6F and Q5F) [4] on selectivity filter gating in TREK-1.



Activators stabilize the selectivity filter in the conductive conformation...



...by two distinct mechanisms



Mendez-Otalvaro, Kopec, de Groot, 'Effect of two activators on the gating of a K2P channel' *bioRxiv* 2024.